CPU Scheduling Simulator

Project Report

A website built with HTML, CSS and JavaScript that simulates nine CPU scheduling algorithms.

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1. Introduction

The CPU scheduler decides which process in the ready queue gets the processor next. The choice affects how long processes wait and how quickly they finish. This project is a website that lets a user enter processes and see how different scheduling algorithms handle them.

For every algorithm the website draws a Gantt chart and calculates the completion time (CT), turnaround time (TAT) and waiting time (WT) of each process, along with their averages.

2. Objectives

- Implement FCFS, SJF, SRTF, LJF, LRTF, Priority (both forms), Round Robin and HRRN in one website.
- Accept process ID, arrival time and burst time, and ask for priority or a time quantum only when the chosen algorithm needs them.
- Show a Gantt chart, a result table and the average CT, TAT and WT.
- Let the user compare all algorithms on the same input.

3. Technologies Used

Technology	File	Role
HTML	index.html	Page structure: left menu, input area and result areas
CSS	style.css	Black and gold theme and responsive layout
JavaScript	script.js	Algorithms, input validation, page switching and calculations

4. Basic Terms and Formulas

Term	Meaning
Arrival time (AT)	The time at which the process enters the ready queue
Burst time (BT)	The CPU time the process needs to complete
Completion time (CT)	The time at which the process finishes
Turnaround time (TAT)	TAT = CT - AT
Waiting time (WT)	WT = TAT - BT
Average	The total of a column divided by the number of processes

5. Scheduling Algorithms

First Come First Serve (FCFS)

Runs processes in the order they arrive. It is simple and never starves a process, but a long job at the front makes every later job wait.

Shortest Job First (SJF)

Among the arrived processes, the one with the smallest burst time runs next and is not interrupted. It gives a low average waiting time, but long jobs may wait a long time.

Shortest Remaining Time First (SRTF)

The preemptive form of SJF. When a new process arrives with a burst shorter than the time the running process has left, it takes the CPU.

Longest Job First (LJF)

Among the arrived processes, the one with the largest burst time runs next and is not interrupted. Long jobs finish early, but short jobs can wait behind them, so the average waiting time is usually high.

Longest Remaining Time First (LRTF)

The preemptive form of LJF. At every time unit the process with the most remaining time runs, so a newly arrived longer job takes the CPU. Processes with equal remaining time take turns.

Priority Scheduling (Non-Preemptive and Preemptive)

The process with the highest priority runs first. In this project a lower number means a higher priority. The non-preemptive form lets the running process finish; the preemptive form lets a higher-priority arrival interrupt it. Low-priority processes can wait indefinitely.

Round Robin (RR)

Every process gets the CPU for a fixed time quantum and then moves to the back of the ready queue if it is not finished. It is fair and responsive, but a small quantum causes many switches and a large one behaves like FCFS.

Highest Response Ratio Next (HRRN)

Picks the process with the highest ratio of (waiting time + burst time) divided by burst time. Short jobs are favoured, but a job that has waited long gains a high ratio and is eventually chosen.

6. System Design

6.1 Pages and navigation

The site has one page for each algorithm, reached from the left menu, plus a page that compares all algorithms. Each page has its own title and description. The process list is kept while the user moves between pages, so the same input can be tried on every algorithm.

6.2 Inputs shown on each page

Page	Inputs
FCFS, SJF, SRTF, LJF, LRTF, HRRN	Process ID, arrival time, burst time
Priority, Priority (Preemptive)	Process ID, arrival time, burst time, priority
Round Robin	Process ID, arrival time, burst time, time quantum
Compare all	All inputs, because every algorithm is run

6.3 How the simulation works

- Non-preemptive algorithms (FCFS, SJF, LJF, Priority, HRRN): when the CPU is free, pick one process from those that have arrived and run it to completion.
- Preemptive algorithms (SRTF, LRTF, Priority Preemptive): advance one time unit at a time and re-pick the best process each unit.
- Round Robin: keep a queue, run the first process for the quantum or until it ends, and add newly arrived processes before re-queuing it.
- If no process has arrived, the CPU is idle until the next arrival, and the Gantt chart shows an Idle block.

6.4 Input checks

The website checks the input before running. It reports a duplicate process ID, an arrival time below 0, a burst time below 1 and a time quantum below 1, and says which process needs fixing.

7. Sample Input and Results

The following input was used to test all nine algorithms. Round Robin used a time quantum of 2.

Process	Arrival time	Burst time	Priority
P1	0	5	2
P2	1	3	1
P3	2	8	4
P4	3	6	3

7.1 FCFS in detail

Gantt chart: P1 (0-5), P2 (5-8), P3 (8-16), P4 (16-22).

Process	AT	BT	CT	TAT	WT
P1	0	5	5	5	0
P2	1	3	8	7	4
P3	2	8	16	14	6
P4	3	6	22	19	13

Average CT = 12.75, average TAT = 11.25, average WT = 5.75.

7.2 Comparison of all algorithms

Algorithm	Avg CT	Avg TAT	Avg WT
FCFS	12.75	11.25	5.75
SJF	12.25	10.75	5.25
SRTF	12	10.50	5
LJF	14.75	13.25	7.75
LRTF	20.50	19	13.50
Priority	12.25	10.75	5.25
Priority (Preemptive)	12	10.50	5
Round Robin	16.75	15.25	9.75
HRRN	12.25	10.75	5.25

The highlighted rows have the lowest average waiting time.

7.3 Observations

- SRTF and Priority (Preemptive) gave the lowest average waiting time (5) for this input.
- LRTF gave the highest (13.50). Longest-first methods make short jobs wait behind long ones, and LRTF also keeps switching between long jobs, so every process finishes late.
- Round Robin (quantum 2) is also high here, because each process waits while the others use their turns.
- FCFS is higher than the shortest-job methods because P3, a long job, runs before the shorter P4.
- The best algorithm depends on the input, which is why the Compare all page is useful.

8. Conclusion

The project delivers a working simulator for nine CPU scheduling algorithms. Each algorithm has its own page with only the inputs it needs, and every run gives a Gantt chart, per-process CT, TAT and WT, and their averages. The comparison page shows how the algorithms differ on the same processes.

9. Future Scope

- Add multilevel queue and multilevel feedback queue scheduling.
- Export the Gantt chart and result table as PDF or CSV.
- Show Gantt charts of several algorithms side by side.
- Include context-switch time in the calculations.